In [ ]:
```python
import marimo as mo
```

VQ-VAE Tokenizer Evaluation

Protein / Ligand VQ-VAE の学習後の定量的性能評価ノートブック。

評価項目:

1. 再構成誤差 (MSE) — 全体 & 次元別
2. Codebook 利用率 & Perplexity
3. Original vs Reconstructed の散布図
4. Codebook 使用頻度の分布
5. 潜在空間の可視化 (t-SNE)

In [ ]:
```python
import sys
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import torch

# Add project root to path (resolve from this file, not cwd)
project_root = Path(__file__).resolve().parent.parent
if str(project_root) not in sys.path:
    sys.path.insert(0, str(project_root))

from src.config import CrossDockedConfig, VQVAETrainingConfig
from src.data.descriptors import ComplexDescriptorDataModule
from src.model.vqvae_module import VQVAEModule

# '%matplotlib inline' command supported automatically in marimo
plt.rcParams["figure.dpi"] = 120
```

1. Load checkpoint and data

In [ ]:
```python
# --- Checkpoint path (edit here) ---
# Best checkpoint is typically the one with lowest val/protein_recon.
# List available checkpoints:
ckpt_dir = project_root / "checkpoints"
if ckpt_dir.exists():
    ckpts = sorted(ckpt_dir.rglob("*.ckpt"))
    for _p in ckpts:
        print(_p.relative_to(project_root))
else:
    print("No checkpoints/ directory found. Check wandb run directory.")
    ckpts = sorted(project_root.rglob("*.ckpt"))
    for _p in ckpts[-5:]:
        print(_p.relative_to(project_root))
```

```
No checkpoints/ directory found. Check wandb run directory.
pocket-ligand-vqvae/uhhyc6y7/checkpoints/vqvae-epoch=86-val/protein_recon=0.
1287.ckpt
pocket-ligand-vqvae/uhhyc6y7/checkpoints/vqvae-epoch=93-val/protein_recon=0.
1288.ckpt
pocket-ligand-vqvae/yggua4f0/checkpoints/vqvae-epoch=90-val/protein_recon=0.
1316.ckpt
pocket-ligand-vqvae/yggua4f0/checkpoints/vqvae-epoch=95-val/protein_recon=0.
1316.ckpt
pocket-ligand-vqvae/yggua4f0/checkpoints/vqvae-epoch=99-val/protein_recon=0.
1316.ckpt
```

In [ ]:
```python
import os

# Override via `VQVAE_CKPT` env var so a single notebook can be exported
# per run without editing source in between. Falls back to the baseline
# (yggua4f0) used in the original evaluation.
CKPT_PATH = os.environ.get(
    "VQVAE_CKPT",
    "/gs/bs/tga-ohuelab/sakano/git/pocket-conditioned-ligand-gen/pocket-liga
)

print(f"Loading: {CKPT_PATH}")

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
module = VQVAEModule.load_from_checkpoint(str(CKPT_PATH), map_location=devic
module.eval()
module.to(device)

protein_vqvae = module.protein_vqvae
ligand_vqvae = module.ligand_vqvae
print(f"Device: {device}")
```

```
Loading: /gs/bs/tga-ohuelab/sakano/git/pocket-conditioned-ligand-gen/pocket-
ligand-vqvae/lv5nldy5/checkpoints/vqvae-epoch=96-val/protein_recon=0.1294.ck
pt
/gs/bs/tga-ohuelab/sakano/git/pocket-conditioned-ligand-gen/.venv/lib/python
3.12/site-packages/torch/nn/modules/transformer.py:392: UserWarning: enable_
nested_tensor is True, but self.use_nested_tensor is False because encoder_l
ayer.norm_first was True
  warnings.warn(
Device: cuda
```

In [ ]:
```python
# Load cached descriptors and prepare test split
config = VQVAETrainingConfig()
data_config = CrossDockedConfig(data_dir=project_root / "data")
dm = ComplexDescriptorDataModule(config, data_config)
dm.setup()

norm_stats = dm.norm_stats

# The full test split is ~2M complexes across 436 shards; loading it all
# takes the better part of an hour and is overkill for MSE / t-SNE
# statistics.  Cap the notebook at MAX_TEST_COMPLEXES by stopping early.
MAX_TEST_COMPLEXES = 2000
```

```python
if dm.protein_test is None:
    prot_mean_np = norm_stats["protein_mean"].numpy()
    prot_std_np = norm_stats["protein_std"].numpy()
    lig_mean_np = norm_stats["ligand_mean"].numpy()
    lig_std_np = norm_stats["ligand_std"].numpy()

    protein_test_pockets = []
    ligand_test_molecules = []
    # Read each test shard once and pull out both protein + ligand,
    # so we don't pay 2x disk I/O across two separate iterators.
    for shard_idx, local_indices in dm._test_plan:
        if len(protein_test_pockets) >= MAX_TEST_COMPLEXES:
            break
        shard_path = dm._shard_dir / f"shard_{shard_idx:04d}.pt"
        shard_data = torch.load(shard_path, weights_only=False)
        for local_idx in local_indices:
            if len(protein_test_pockets) >= MAX_TEST_COMPLEXES:
                break
            cplx = shard_data[local_idx]
            protein_test_pockets.append(
                torch.from_numpy(
                    (cplx["protein"] - prot_mean_np) / prot_std_np,
                )
                .float()
                .to(device),
            )
            ligand_test_molecules.append(
                torch.from_numpy(
                    (cplx["ligand"] - lig_mean_np) / lig_std_np,
                )
                .float()
                .to(device),
            )
        del shard_data
else:
    protein_test_pockets = [
        t.to(device) for t in dm.protein_test[:MAX_TEST_COMPLEXES]
    ]
    ligand_test_molecules = [
        t.to(device) for t in dm.ligand_test[:MAX_TEST_COMPLEXES]
    ]

protein_test_flat = torch.cat(protein_test_pockets)
ligand_test_flat = torch.cat(ligand_test_molecules)

print(
    f"Protein test: {len(protein_test_pockets)} pockets, "
    f"{protein_test_flat.shape[0]} residues total"
)
print(
    f"Ligand test:  {len(ligand_test_molecules)} molecules, "
    f"{ligand_test_flat.shape[0]} atoms total"
)
```

```
Protein test: 2000 pockets, 55427 residues total
Ligand test:  2000 molecules, 56302 atoms total
```

In [ ]:
```python
# Both protein and ligand share the same TransformerVQVAE architecture and
# consume variable-length per-pocket / per-molecule sequences.
@torch.no_grad()
def run_vqvae(model: "TransformerVQVAE", sequences: "list[Tensor]"):
    """Run TransformerVQVAE per-sequence and collect recon/indices/z."""
    all_recon, all_indices, all_z = [], [], []
    for seq in sequences:
        if seq.shape[0] == 0:
            continue
        x_seq = seq.unsqueeze(0)  # (1, N, D)
        h = model.input_proj(model.input_norm(x_seq))
        h = h + model.pos_encoding[: seq.shape[0]]
        h = model.transformer_encoder(h)
        z = model.latent_norm(model.latent_proj(h)).squeeze(0)  # (N, latent
        quantized, indices, _, _ = model.codebook(z)
        q_seq = model.latent_unproj(quantized).unsqueeze(0)
        dec_in = q_seq + model.pos_encoding[: seq.shape[0]]
        dec_out = model.transformer_decoder(dec_in)
        recon = model.output_proj(dec_out).squeeze(0)  # (N, D)
        all_recon.append(recon)
        all_indices.append(indices)
        all_z.append(z)
    return torch.cat(all_recon), torch.cat(all_indices), torch.cat(all_z)

prot_recon, prot_indices, prot_z = run_vqvae(protein_vqvae, protein_test_poc
lig_recon, lig_indices, lig_z = run_vqvae(ligand_vqvae, ligand_test_molecule

print("Inference done.")
print(
    f"Protein: recon {prot_recon.shape}, indices {prot_indices.shape}, z {pr
)
print(
    f"Ligand:  recon {lig_recon.shape}, indices {lig_indices.shape}, z {lig_
)
```

```
Inference done.
Protein: recon torch.Size([55427, 12]), indices torch.Size([55427]), z torc
h.Size([55427, 16])
Ligand:  recon torch.Size([56302, 4]), indices torch.Size([56302]), z torch.
Size([56302, 8])
```

2. Reconstruction Error (MSE)

正規化空間と元スケールの両方で再構成誤差を評価する。

In [ ]:
```python
def compute_mse_metrics(
    original: "Tensor",
    reconstructed: "Tensor",
    name: str,
    norm_mean: "Tensor",
    norm_std: "Tensor",
):
    """Compute MSE in both normalized and original scale."""
    orig_np = original.cpu().numpy()
```

```
    recon_np = reconstructed.cpu().numpy()

    # Normalized space
    mse_norm = np.mean((orig_np - recon_np) ** 2)
    per_dim_mse_norm = np.mean((orig_np - recon_np) ** 2, axis=0)

    # De-normalize to original scale
    mean_np = norm_mean.numpy()
    std_np = norm_std.numpy()
    orig_denorm = orig_np * std_np + mean_np
    recon_denorm = recon_np * std_np + mean_np
    mse_orig = np.mean((orig_denorm - recon_denorm) ** 2)
    per_dim_mse_orig = np.mean((orig_denorm - recon_denorm) ** 2, axis=0)

    print(f"=== {name} VQ-VAE ===")
    print(f"  Overall MSE (normalized):  {mse_norm:.6f}")
    print(f"  Overall MSE (original):    {mse_orig:.6f}")
    print(f"  Overall RMSE (original):   {np.sqrt(mse_orig):.6f}")
    print()

    return {
        "mse_norm": mse_norm,
        "per_dim_mse_norm": per_dim_mse_norm,
        "mse_orig": mse_orig,
        "per_dim_mse_orig": per_dim_mse_orig,
        "orig_denorm": orig_denorm,
        "recon_denorm": recon_denorm,
    }

prot_metrics = compute_mse_metrics(
    protein_test_flat,
    prot_recon,
    "Protein",
    norm_stats["protein_mean"],
    norm_stats["protein_std"],
)
lig_metrics = compute_mse_metrics(
    ligand_test_flat,
    lig_recon,
    "Ligand",
    norm_stats["ligand_mean"],
    norm_stats["ligand_std"],
)
```

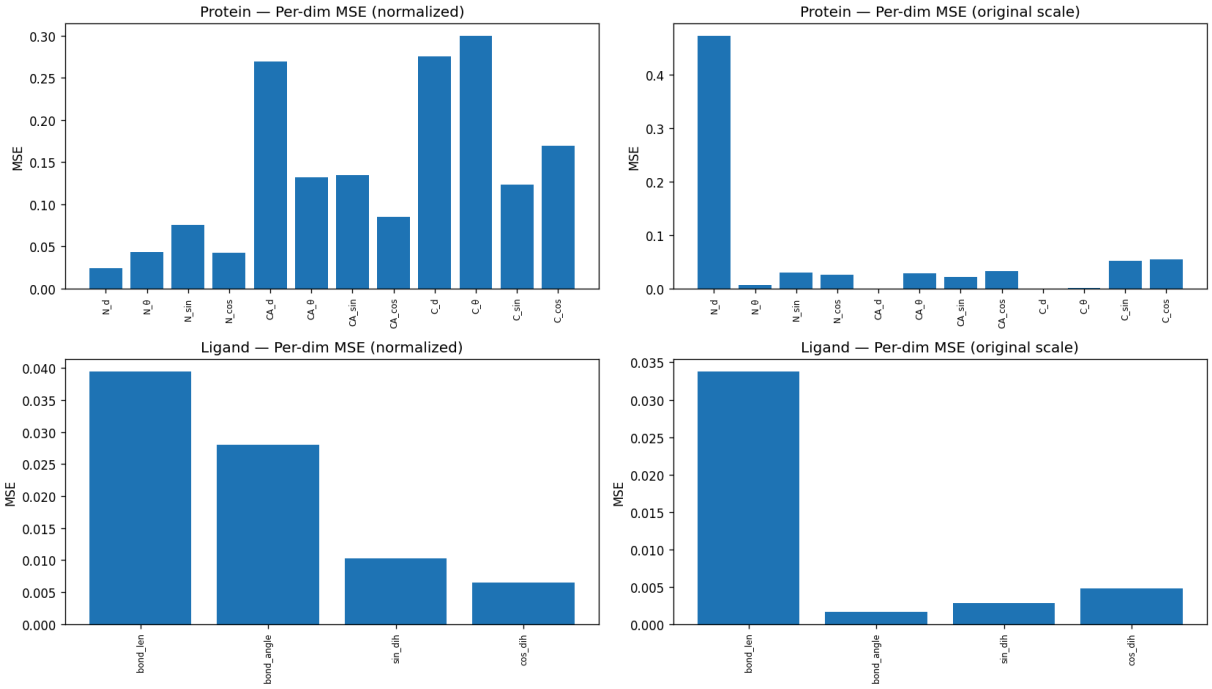
```
=== Protein VQ-VAE ===
  Overall MSE (normalized):  0.139450
  Overall MSE (original):    0.060466
  Overall RMSE (original):   0.245899

=== Ligand VQ-VAE ===
  Overall MSE (normalized):  0.021067
  Overall MSE (original):    0.010741
  Overall RMSE (original):   0.103641
```

2.1 Per-dimension MSE

Protein: 12-D backbone Z-matrix (4 values × 3 atoms N/CA/C). 4 values per atom are `(bond_length, bond_angle, sin_torsion, cos_torsion)` for continuation residues, or pocket-frame-anchored spherical coords for segment-start residues. Ligand: dim 0 = bond length, dim 1 = bond angle, dim 2–3 = sin/cos dihedral

In [ ]:

```
_fig, _axes = plt.subplots(2, 2, figsize=(14, 8))
prot_dim_labels = [
    "N_d",
    "N_θ",
    "N_sin",
    "N_cos",
    "CA_d",
    "CA_θ",
    "CA_sin",
    "CA_cos",
    "C_d",
    "C_θ",
    "C_sin",
    "C_cos",
]
n_prot_dims = len(prot_dim_labels)
# Protein — normalized
_axes[0, 0].bar(range(n_prot_dims), prot_metrics["per_dim_mse_norm"])
_axes[0, 0].set_xticks(range(n_prot_dims), prot_dim_labels, rotation=90, for
_axes[0, 0].set_title("Protein — Per-dim MSE (normalized)")
_axes[0, 0].set_ylabel("MSE")
_axes[0, 1].bar(range(n_prot_dims), prot_metrics["per_dim_mse_orig"])
_axes[0, 1].set_xticks(range(n_prot_dims), prot_dim_labels, rotation=90, for
# Protein — original scale
_axes[0, 1].set_title("Protein — Per-dim MSE (original scale)")
_axes[0, 1].set_ylabel("MSE")
lig_dim_labels = ["bond_len", "bond_angle", "sin_dih", "cos_dih"]
_axes[1, 0].bar(range(4), lig_metrics["per_dim_mse_norm"])
_axes[1, 0].set_xticks(range(4), lig_dim_labels, rotation=90, fontsize=7)
# Ligand — normalized
_axes[1, 0].set_title("Ligand — Per-dim MSE (normalized)")
_axes[1, 0].set_ylabel("MSE")
_axes[1, 1].bar(range(4), lig_metrics["per_dim_mse_orig"])
_axes[1, 1].set_xticks(range(4), lig_dim_labels, rotation=90, fontsize=7)
_axes[1, 1].set_title("Ligand — Per-dim MSE (original scale)")
_axes[1, 1].set_ylabel("MSE")
# Ligand — original scale
_fig.tight_layout()
_fig
```



3. Codebook Utilization & Perplexity

理想的なcodebookは全コードが均等に使われる。

- **Utilization**: 使用されたコード数 / 全コード数 (1.0が理想)
- **Perplexity**: exp(entropy)。均等利用時は codebook_size と一致

In [ ]:

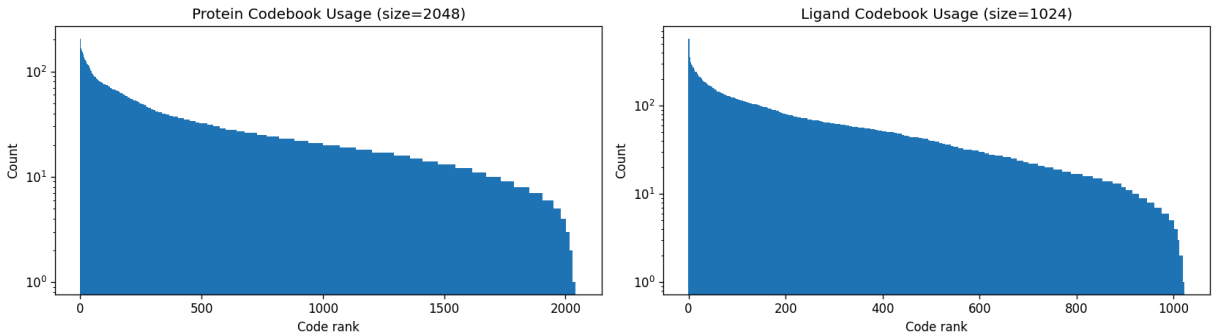
```python
def codebook_stats(indices: "Tensor", codebook_size: int, name: str):
    """Compute and print codebook utilization metrics."""
    idx_np = indices.cpu().numpy()
    unique = np.unique(idx_np)
    utilization = len(unique) / codebook_size

    counts = np.bincount(idx_np, minlength=codebook_size).astype(float)
    probs = counts / counts.sum()
    probs_nonzero = probs[probs > 0]
    entropy = -np.sum(probs_nonzero * np.log(probs_nonzero))
    perplexity = np.exp(entropy)

    # Max perplexity = codebook_size (uniform distribution)
    max_perplexity = codebook_size
    normalized_perplexity = perplexity / max_perplexity

    print(f"=== {name} Codebook ===")
    print(f"  Codebook size:          {codebook_size}")
    print(f"  Active codes:           {len(unique)}")
    print(f"  Utilization:            {utilization:.4f}")
    print(f"  Perplexity:             {perplexity:.1f} / {max_perplexity}")
    print(f"  Normalized perplexity:  {normalized_perplexity:.4f}")
    print(f"  Dead codes:             {codebook_size - len(unique)}")
    print()
    return counts
```

```
prot_counts = codebook_stats(prot_indices, config.protein.codebook_size, "Pr
lig_counts = codebook_stats(lig_indices, config.ligand.codebook_size, "Ligar
```

```
=== Protein Codebook ===
  Codebook size:          2048
  Active codes:           2042
  Utilization:            0.9971
  Perplexity:             1517.8 / 2048
  Normalized perplexity:  0.7411
  Dead codes:             6

=== Ligand Codebook ===
  Codebook size:          1024
  Active codes:           1023
  Utilization:            0.9990
  Perplexity:             694.8 / 1024
  Normalized perplexity:  0.6785
  Dead codes:             1
```

3.1 Codebook usage distribution

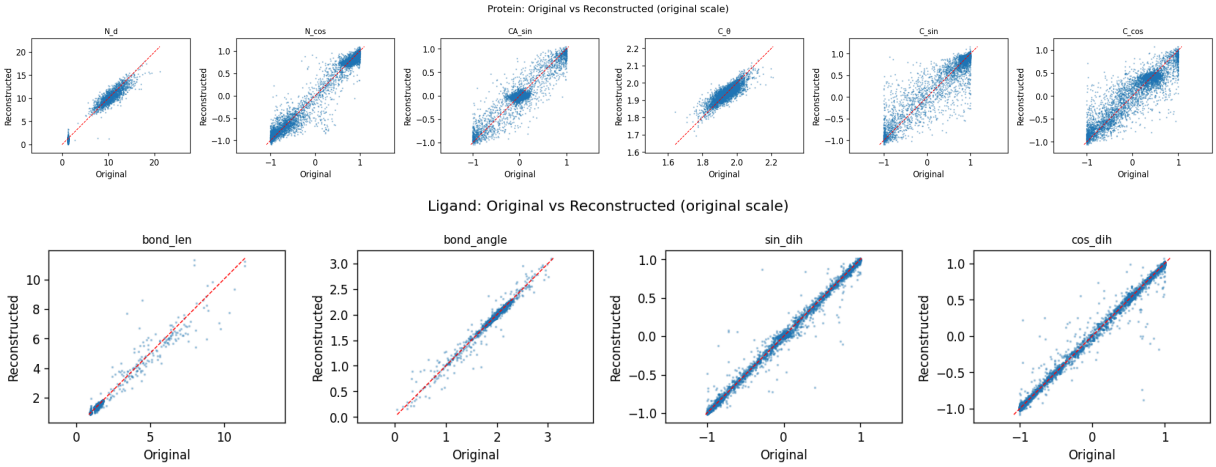
In [ ]:
```
_fig, _axes = plt.subplots(1, 2, figsize=(14, 4))
prot_sorted = np.sort(prot_counts)[::-1]
# Protein codebook usage — sorted descending
_axes[0].bar(range(len(prot_sorted)), prot_sorted, width=1.0)
_axes[0].set_xlabel("Code rank")
_axes[0].set_ylabel("Count")
_axes[0].set_title(f"Protein Codebook Usage (size={config.protein.codebook_s
_axes[0].set_yscale("log")
lig_sorted = np.sort(lig_counts)[::-1]
_axes[1].bar(range(len(lig_sorted)), lig_sorted, width=1.0)
# Ligand codebook usage — sorted descending
_axes[1].set_xlabel("Code rank")
_axes[1].set_ylabel("Count")
_axes[1].set_title(f"Ligand Codebook Usage (size={config.ligand.codebook_siz
_axes[1].set_yscale("log")
_fig.tight_layout()
_fig
```



4. Original vs Reconstructed scatter plots

各次元で元の値 vs 再構成値をプロットし、対角線からのずれを確認する。元スケールで表示。

In [ ]:
```python
def scatter_orig_vs_recon(
    orig: "ndarray",
    recon: "ndarray",
    dim_labels: list[str],
    title: str,
    max_points: int = 5000,
):
    """Scatter plot of original vs reconstructed for selected dimensions."""
    n_dims = orig.shape[1]
    max_inline_dims = 8
    if n_dims <= max_inline_dims:
        selected = list(range(n_dims))
    else:
        selected = [
            0,
            n_dims // 4,
            n_dims // 2,
            3 * n_dims // 4,
            n_dims - 2,
            n_dims - 1,
        ]
    n_plots = len(selected)
    _fig, _axes = plt.subplots(1, n_plots, figsize=(3.5 * n_plots, 3.2))
    if n_plots == 1:
        _axes = [_axes]
    rng = np.random.default_rng(42)
    idx = rng.choice(len(orig), min(max_points, len(orig)), replace=False)
    for ax, dim in zip(_axes, selected, strict=False):
        ax.scatter(orig[idx, dim], recon[idx, dim], s=1, alpha=0.3)
        lo = min(orig[idx, dim].min(), recon[idx, dim].min())
        hi = max(orig[idx, dim].max(), recon[idx, dim].max())
        ax.plot([lo, hi], [lo, hi], "r--", linewidth=0.8)
        ax.set_xlabel("Original")
        ax.set_ylabel("Reconstructed")
        ax.set_title(dim_labels[dim], fontsize=9)
        ax.set_aspect("equal", adjustable="datalim")  # Subsample for plotti
    _fig.suptitle(title, fontsize=12)
    _fig.tight_layout()
    plt.show()

scatter_orig_vs_recon(
    prot_metrics["orig_denorm"],
    prot_metrics["recon_denorm"],
    prot_dim_labels,
    "Protein: Original vs Reconstructed (original scale)",
)
scatter_orig_vs_recon(
    lig_metrics["orig_denorm"],
    lig_metrics["recon_denorm"],
    lig_dim_labels,
    "Ligand: Original vs Reconstructed (original scale)",
)
```



5. Reconstruction error distribution

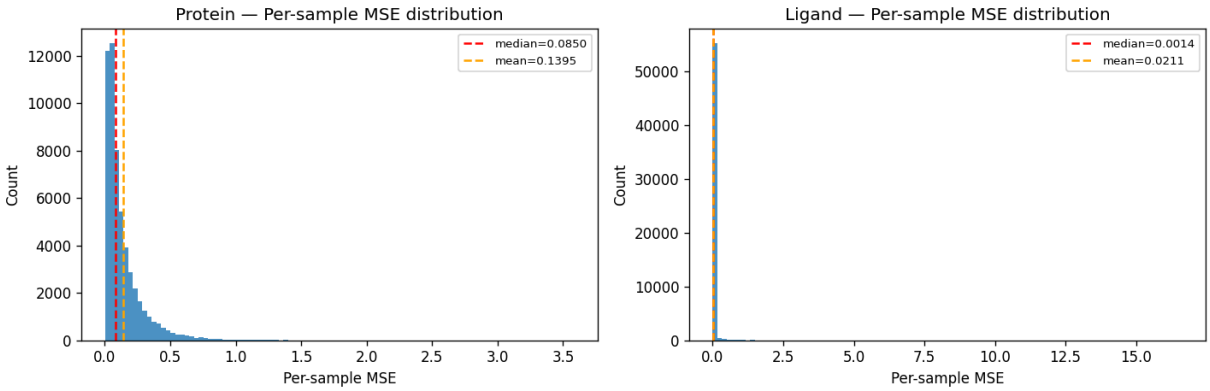
サンプルごとの再構成誤差の分布を確認し、外れ値の存在を把握する。

In [ ]:

```
_fig, _axes = plt.subplots(1, 2, figsize=(12, 4))
prot_per_sample = np.mean(
    (protein_test_flat.cpu().numpy() - prot_recon.cpu().numpy()) ** 2, axis=
)
lig_per_sample = np.mean(
    (ligand_test_flat.cpu().numpy() - lig_recon.cpu().numpy()) ** 2, axis=1
)
_axes[0].hist(prot_per_sample, bins=100, edgecolor="none", alpha=0.8)
_axes[0].axvline(
    np.median(prot_per_sample),
    color="r",
    linestyle="--",
    label=f"median={np.median(prot_per_sample):.4f}",
)
_axes[0].axvline(
    np.mean(prot_per_sample),
    color="orange",
    linestyle="--",
    label=f"mean={np.mean(prot_per_sample):.4f}",
)
_axes[0].set_xlabel("Per-sample MSE")
_axes[0].set_ylabel("Count")
_axes[0].set_title("Protein — Per-sample MSE distribution")
_axes[0].legend(fontsize=8)
_axes[1].hist(lig_per_sample, bins=100, edgecolor="none", alpha=0.8)
_axes[1].axvline(
    np.median(lig_per_sample),
    color="r",
    linestyle="--",
    label=f"median={np.median(lig_per_sample):.4f}",
)
_axes[1].axvline(
    np.mean(lig_per_sample),
    color="orange",
    linestyle="--",
```

```
    label=f"mean={np.mean(lig_per_sample):.4f}",
)
_axes[1].set_xlabel("Per-sample MSE")
_axes[1].set_ylabel("Count")
_axes[1].set_title("Ligand — Per-sample MSE distribution")
_axes[1].legend(fontsize=8)
_fig.tight_layout()
for name, mse_arr in [("Protein", prot_per_sample), ("Ligand", lig_per_sampl
    print(f"{name} per-sample MSE percentiles:")
    for _p in [50, 75, 90, 95, 99]:
        print(f"  P{_p}: {np.percentile(mse_arr, _p):.6f}")
    print()
_fig
```

```
Protein per-sample MSE percentiles:
  P50: 0.084992
  P75: 0.175036
  P90: 0.318315
  P95: 0.442415
  P99: 0.781700

Ligand per-sample MSE percentiles:
  P50: 0.001380
  P75: 0.003945
  P90: 0.010937
  P95: 0.026193
  P99: 0.421767
```



5.5 3D Reconstruction RMSD (Å)

記述子空間での再構成誤差ではなく、実際の3次元座標に復元した上での RMSD を評価する。

- **Protein**: backbone (N, CA, C) 座標の RMSD
- **Ligand**: 重原子座標の RMSD

2 種類の RMSD を並べて報告する:

- **Per-atom (superposition なし)**: pocket frame を共有しているため、segment-start のズレや NeRF 累積による剛体シフトがそのまま出る。生成 pipeline 全体の精

度。

- **Kabsch-aligned**: 原子集合を剛体 (回転 + 並進) でベストフィットさせた後の RMSD。内部変形 (conformer shape) の再現精度のみを反映する。

複数の複合体をサンプリングし、VQ-VAE の encode → decode → 記述子逆変換 → 3D座標 のパイプライン全体の精度を確認する。

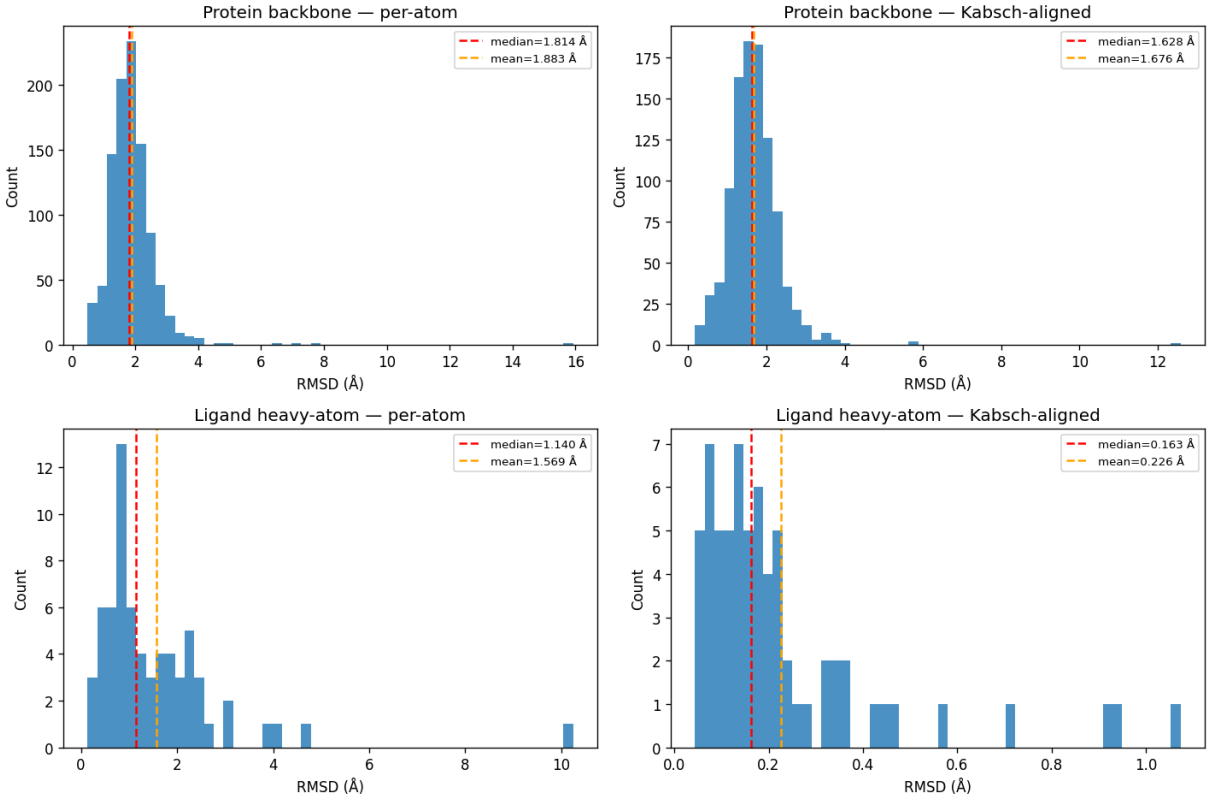
In [ ]:
```python
import pyarrow.parquet as pq

from src.config import PocketExtractionConfig
from src.tokenizers.ligand import LigandDescriptor, parse_sdf
from src.tokenizers.protein import (
    BackboneZMatrixDescriptor,
    _compute_canonical_frame,
    extract_pocket,
)

def kabsch_rmsd(p: np.ndarray, q: np.ndarray) -> float:
    """Per-atom RMSD after optimal rigid-body alignment of q onto p (Kabsch)

    Both inputs shape (N, 3). Returns RMSD in the same units.
    """
    p_c = p - p.mean(axis=0)
    q_c = q - q.mean(axis=0)
    h = q_c.T @ p_c
    u, _, vt = np.linalg.svd(h)
    d = np.sign(np.linalg.det(vt.T @ u.T))
    rot = vt.T @ np.diag([1.0, 1.0, d]) @ u.T
    q_aligned = q_c @ rot.T
    return float(np.sqrt(np.mean(np.sum((p_c - q_aligned) ** 2, axis=-1))))

pocket_config = PocketExtractionConfig()
protein_desc_calc = BackboneZMatrixDescriptor()
ligand_desc_calc = LigandDescriptor()
# Use the HuggingFace hub cache manifest instead of the CrossDocked2020
# types/ tree (which isn't present on this machine).
hub_cache_dir = project_root / "data" / "hub_cache"
manifest_path = hub_cache_dir / "repo" / "manifest.parquet"
receptor_dir = hub_cache_dir / "receptors"
ligand_dir = hub_cache_dir / "ligands"
manifest_df = pq.read_table(manifest_path).to_pandas()
# Filter to the cdonly test split for fold 0 — matches the original
# cdonly_*_test0.types file that this cell used to read.
test_df = manifest_df[
    (manifest_df["source_type"] == "cdonly")
    & (manifest_df["cdonly_fold0"] == "test")
].reset_index(drop=True)
entries = [
    (
        f"{row.complex_dir}/{row.receptor_pdb}",
        f"{row.pair_idx:07d}.sdf.gz",
        0,
    )
    for row in test_df.itertuples(index=False)
```

```
]
prot_mean = norm_stats["protein_mean"].to(device)
prot_std = norm_stats["protein_std"].to(device)
lig_mean = norm_stats["ligand_mean"].to(device)
lig_std = norm_stats["ligand_std"].to(device)
N_SAMPLES = 200
rng = np.random.default_rng(42)
sample_indices = rng.choice(
    len(entries), size=min(N_SAMPLES * 5, len(entries)), replace=False
)
prot_rmsd_list = []
prot_rmsd_aligned_list = []
lig_rmsd_list = []
lig_rmsd_aligned_list = []
n_done = 0
for idx in sample_indices:
    if n_done >= N_SAMPLES:
        break
    rec_rel, lig_rel, _pose_idx = entries[idx]
    rec_path = receptor_dir / rec_rel
    lig_path = ligand_dir / lig_rel
    if not rec_path.exists() or not lig_path.exists():
        continue
    try:
        molecules = parse_sdf(lig_path)
        if not molecules:
            continue
        mol = molecules[0]
        lig_coords_orig = np.array(
            [(a[1], a[2], a[3]) for a in mol["atoms"] if a[0] != "H"],
            dtype=np.float32,
        )
        pocket_result = extract_pocket(rec_path, lig_coords_orig, pocket_cor
        if pocket_result is None:
            continue
        backbone_coords_orig, _seq, residue_ids = pocket_result
        ca_coords = backbone_coords_orig[:, 1].astype(np.float64)
        centroid, rotation = _compute_canonical_frame(ca_coords)
        pocket_frame = (centroid, rotation)
        prot_desc, prot_meta = protein_desc_calc.compute(
            backbone_coords_orig,
            residue_ids,
            pocket_frame=pocket_frame,
        )
        prot_t = torch.from_numpy(prot_desc).to(device)
        prot_norm = (prot_t - prot_mean) / prot_std
        with torch.no_grad():
            pi = protein_vqvae.encode(prot_norm)
            prot_recon_norm = protein_vqvae.decode(pi)
        prot_recon_desc = (prot_recon_norm * prot_std + prot_mean).cpu().num
        backbone_recon = BackboneZMatrixDescriptor.descriptor_to_backbone_cc
            prot_recon_desc, prot_meta
        )
        prot_rmsd = np.sqrt(
            np.mean(np.sum((backbone_coords_orig - backbone_recon) ** 2, axi
        )
```

```
        prot_rmsd_list.append(prot_rmsd)
        prot_flat_orig = backbone_coords_orig.reshape(-1, 3).astype(np.float
        prot_flat_recon = backbone_recon.reshape(-1, 3).astype(np.float64)
        prot_rmsd_aligned_list.append(
            kabsch_rmsd(prot_flat_orig, prot_flat_recon)
        )
        lig_desc, _elements, lig_meta = ligand_desc_calc.compute(
            mol["atoms"], mol["bonds"], pocket_frame=pocket_frame
        )
        if len(lig_desc) == 0:
            continue
        lig_t = torch.from_numpy(lig_desc).to(device)
        lig_norm = (lig_t - lig_mean) / lig_std
        with torch.no_grad():
            li = ligand_vqvae.encode(lig_norm)
            lig_recon_norm = ligand_vqvae.decode(li)
        lig_recon_desc = (lig_recon_norm * lig_std + lig_mean).cpu().numpy()
        lig_coords_recon = LigandDescriptor.descriptor_to_coords(
            lig_recon_desc, lig_meta, pocket_frame=pocket_frame
        )
        heavy_atoms = [a for a in mol["atoms"] if a[0] != "H"]
        lig_coords_orig_arr = np.array(
            [(a[1], a[2], a[3]) for a in heavy_atoms], dtype=np.float64
        )
        lig_rmsd = np.sqrt(
            np.mean(
                np.sum((lig_coords_orig_arr - lig_coords_recon) ** 2, axis=-
            )
        )
        lig_rmsd_list.append(lig_rmsd)
        lig_rmsd_aligned_list.append(
            kabsch_rmsd(lig_coords_orig_arr, lig_coords_recon.astype(np.floa
        )
        n_done += 1
    except Exception:  # noqa: BLE001, S112
        continue
prot_rmsd_arr = np.array(prot_rmsd_list)
prot_rmsd_aligned_arr = np.array(prot_rmsd_aligned_list)
lig_rmsd_arr = np.array(lig_rmsd_list)
lig_rmsd_aligned_arr = np.array(lig_rmsd_aligned_list)
print(f"Evaluated {len(prot_rmsd_arr)} complexes")
print()

def _print_rmsd_stats(name: str, arr: np.ndarray) -> None:
    print(f"{name}:")
    print(f"  Mean:   {arr.mean():.4f}")
    print(f"  Median: {np.median(arr):.4f}")
    print(f"  Std:    {arr.std():.4f}")

_print_rmsd_stats("Protein backbone RMSD — per-atom (Å)", prot_rmsd_arr)
print()
_print_rmsd_stats(
    "Protein backbone RMSD — Kabsch-aligned (Å)", prot_rmsd_aligned_arr
)
print()
_print_rmsd_stats("Ligand heavy-atom RMSD — per-atom (Å)", lig_rmsd_arr)
```

```
print()
_print_rmsd_stats(
    "Ligand heavy-atom RMSD — Kabsch-aligned (Å)", lig_rmsd_aligned_arr
)

def _plot_rmsd_hist(ax, arr: np.ndarray, title: str) -> None:
    ax.hist(arr, bins=50, edgecolor="none", alpha=0.8)
    ax.axvline(
        np.median(arr),
        color="r",
        linestyle="--",
        label=f"median={np.median(arr):.3f} Å",
    )
    ax.axvline(
        arr.mean(),
        color="orange",
        linestyle="--",
        label=f"mean={arr.mean():.3f} Å",
    )
    ax.set_xlabel("RMSD (Å)")
    ax.set_ylabel("Count")
    ax.set_title(title)
    ax.legend(fontsize=8)

_fig, _axes = plt.subplots(2, 2, figsize=(12, 8))
_plot_rmsd_hist(_axes[0, 0], prot_rmsd_arr, "Protein backbone — per-atom")
_plot_rmsd_hist(
    _axes[0, 1], prot_rmsd_aligned_arr, "Protein backbone — Kabsch-aligned"
)
_plot_rmsd_hist(_axes[1, 0], lig_rmsd_arr, "Ligand heavy-atom — per-atom")
_plot_rmsd_hist(
    _axes[1, 1], lig_rmsd_aligned_arr, "Ligand heavy-atom — Kabsch-aligned"
)
_fig.tight_layout()
_fig
```

```
Evaluated 998 complexes

Protein backbone RMSD — per-atom (Å):
  Mean:   1.8827
  Median: 1.8140
  Std:    0.7977

Protein backbone RMSD — Kabsch-aligned (Å):
  Mean:   1.6760
  Median: 1.6280
  Std:    0.6939

Ligand heavy-atom RMSD — per-atom (Å):
  Mean:   1.5686
  Median: 1.1405
  Std:    1.4141

Ligand heavy-atom RMSD — Kabsch-aligned (Å):
  Mean:   0.2260
  Median: 0.1635
  Std:    0.2057
```



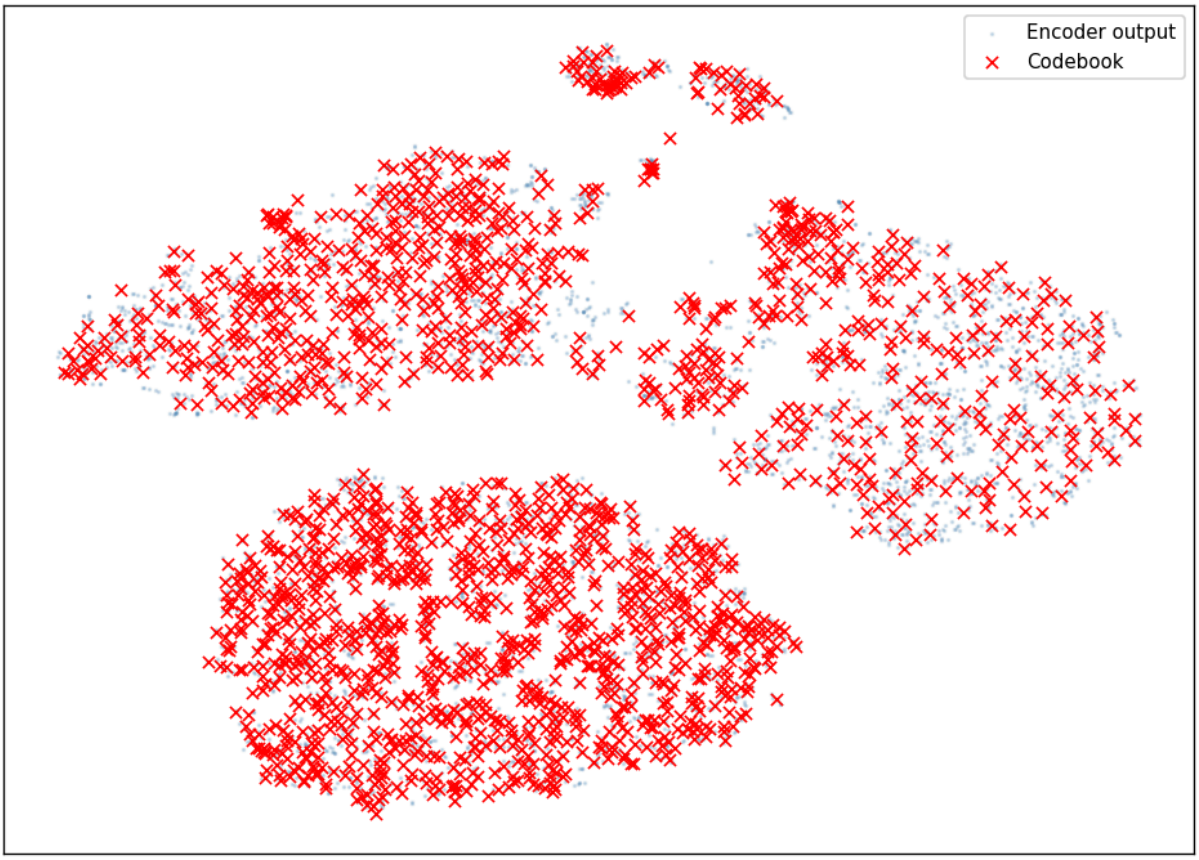
6. Latent space visualization (t-SNE)

Codebook ベクトルとエンコーダ出力を t-SNE で2次元に射影し、量子化の質を確認する。

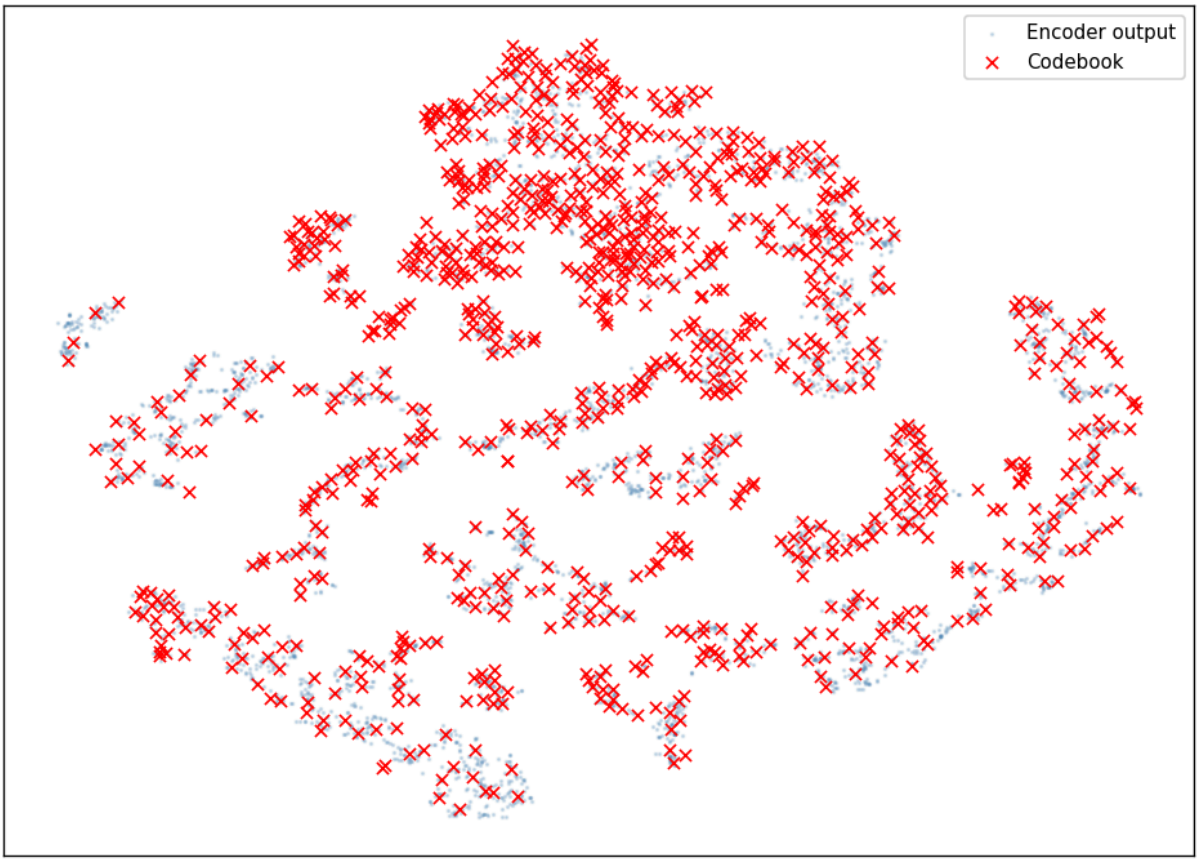
In [ ]:
```python
from sklearn.manifold import TSNE

def plot_latent_tsne(
```

```
    encoder_z: "Tensor",
    codebook: "EMACodebook",
    _codebook_size: int,
    name: str,
    n_samples: int = 5000,
) -> None:
    """t-SNE visualization of encoder outputs and codebook vectors.

    Vectors are plotted as-is; the codebook does plain L2 nearest-neighbor
    lookup on raw embeddings (encoder z is post-LayerNorm).
    """
    z_arr = encoder_z.cpu().numpy()
    cb_arr = codebook.embedding.cpu().detach().numpy()
    rng = np.random.default_rng(42)
    idx = rng.choice(len(z_arr), min(n_samples, len(z_arr)), replace=False)
    z_sub = z_arr[idx]
    combined = np.vstack([z_sub, cb_arr])  # Subsample encoder outputs
    tsne = TSNE(
        n_components=2, random_state=42, perplexity=min(30, len(combined) //
    )
    embedded = tsne.fit_transform(combined)
    z_emb = embedded[: len(z_sub)]
    cb_emb = embedded[len(z_sub) :]
    _fig, ax = plt.subplots(figsize=(8, 6))  # Combine for t-SNE
    ax.scatter(
        z_emb[:, 0],
        z_emb[:, 1],
        s=1,
        alpha=0.2,
        c="steelblue",
        label="Encoder output",
    )
    ax.scatter(
        cb_emb[:, 0],
        cb_emb[:, 1],
        s=30,
        c="red",
        marker="x",
        linewidths=1,
        label="Codebook",
    )
    ax.set_title(f"{name} — t-SNE of latent space")
    ax.legend(fontsize=9)
    ax.set_xticks([])
    ax.set_yticks([])
    _fig.tight_layout()
    plt.show()

plot_latent_tsne(
    prot_z, protein_vqvae.codebook, config.protein.codebook_size, "Protein"
)
plot_latent_tsne(
    lig_z, ligand_vqvae.codebook, config.ligand.codebook_size, "Ligand"
)
```

Protein — t-SNE of latent space



Ligand — t-SNE of latent space



7. Summary table

In [ ]:

```
import pandas as pd

prot_idx_np = prot_indices.cpu().numpy()
lig_idx_np = lig_indices.cpu().numpy()

prot_probs = np.bincount(
    prot_idx_np, minlength=config.protein.codebook_size
).astype(float)
prot_probs = prot_probs / prot_probs.sum()
prot_probs_nz = prot_probs[prot_probs > 0]
prot_ppl = np.exp(-np.sum(prot_probs_nz * np.log(prot_probs_nz)))

lig_probs = np.bincount(lig_idx_np, minlength=config.ligand.codebook_size).a
    float
)
lig_probs = lig_probs / lig_probs.sum()
lig_probs_nz = lig_probs[lig_probs > 0]
lig_ppl = np.exp(-np.sum(lig_probs_nz * np.log(lig_probs_nz)))

summary = pd.DataFrame(
    {
        "Metric": [
            "Codebook size",
            "Latent dim",
            "Test MSE (normalized)",
            "Test MSE (original scale)",
            "Test RMSE (original scale)",
            "Active codes",
            "Utilization",
            "Perplexity",
            "Perplexity (normalized)",
        ],
        "Protein": [
            config.protein.codebook_size,
            config.protein.latent_dim,
            f"{prot_metrics['mse_norm']:.6f}",
            f"{prot_metrics['mse_orig']:.6f}",
            f"{np.sqrt(prot_metrics['mse_orig']):.6f}",
            len(np.unique(prot_idx_np)),
            f"{len(np.unique(prot_idx_np)) / config.protein.codebook_size:.4
            f"{prot_ppl:.1f}",
            f"{prot_ppl / config.protein.codebook_size:.4f}",
        ],
        "Ligand": [
            config.ligand.codebook_size,
            config.ligand.latent_dim,
            f"{lig_metrics['mse_norm']:.6f}",
            f"{lig_metrics['mse_orig']:.6f}",
            f"{np.sqrt(lig_metrics['mse_orig']):.6f}",
            len(np.unique(lig_idx_np)),
            f"{len(np.unique(lig_idx_np)) / config.ligand.codebook_size:.4f}
            f"{lig_ppl:.1f}",
```

```
            f"{lig_ppl / config.ligand.codebook_size:.4f}",
        ],
    }
)

summary
```

	Metric	Protein	Ligand
0	Codebook size	2048	1024
1	Latent dim	16	8
2	Test MSE (normalized)	0.139450	0.021067
3	Test MSE (original scale)	0.060466	0.010741
4	Test RMSE (original scale)	0.245899	0.103641
5	Active codes	2042	1023
6	Utilization	0.9971	0.9990
7	Perplexity	1517.8	694.8
8	Perplexity (normalized)	0.7411	0.6785